

FitTrack: AI-Powered Exercise Tracking Using Computer Vision and Deep Learning

Bachelor's Thesis Presentation

Pătrașca Andrei Ioan

West University of Timișoara
Faculty of Mathematics and Informatics
Informatics in English
Coordinator: Drăgan Ioan

June 30, 2026



The Problem



Why exercise tracking is harder than it looks

Common fitness tracking challenges:

- People often **miscount** repetitions when exercising alone
- Most visual exercise counter apps support **only one** exercise type, not multiple
- Personal trainers are **expensive** and not always available
- Existing apps require **manual input** — users forget or lie
- Wearables track motion but **cannot identify** exercise type

The Core Question

“Can a computer vision system automatically detect, classify, and count fitness exercises in real time — using only a standard webcam?”

Target exercises

 Push-ups  Squats  Bicep Curls

Computer Vision approaches:

- **OpenPose** (CMU, 2017) — first real-time multi-person pose estimation; computationally expensive
- **MediaPipe Pose** (Google, 2020) — 33 landmarks, fast on CPU; limited to body-only tasks
- **YOLOv8-Pose** (Ultralytics, 2023) — state-of-the-art real-time pose estimation; 17 COCO keypoints, GPU-optimized

Fitness applications:

- **Apple Fitness+** — uses Apple Watch accelerometers; no vision-based rep counting
- **Kemtai** — web-based pose tracking; subscription required, no offline mode
- **RepCount dataset** (2022) — benchmark for repetition counting; focuses on video-level counting, not real-time

Feature Comparison

Existing fitness applications vs. FitTrack

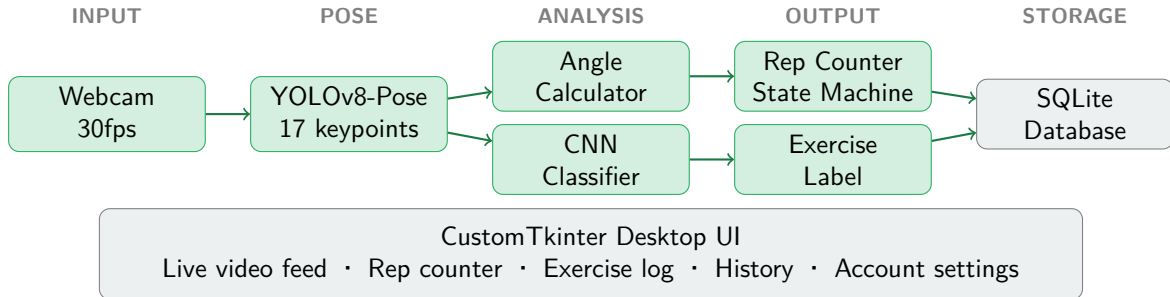


Feature	Apple Fitness+	Kemtai	RepCount	FitTrack
Vision-based	No	Yes	Yes	Yes
Real-time	Yes	Yes	No	Yes
Auto-identification	No	No	No	Yes (CNN)
Multi-exercise	Yes	Yes	Yes	Yes
Offline	No	No	N/A	Yes
Free	No	No	Yes	Yes
Platform	P	W	P	P

Platform: **A** = Android, **P** = PC, **W** = Web

System Architecture

Four-layer pipeline



Pose Estimation

- YOLOv8n-Pose returns 17 joint keypoints per frame
- Only body keypoints drawn — no face clutter

Joint Angle Calculation

- Angle at joint B between points A , B , C :

$$\theta = \arccos\left(\frac{\vec{BA} \cdot \vec{BC}}{|\vec{BA}| |\vec{BC}|}\right)$$

- Elbow $\rightarrow$ push-ups & curls, knee $\rightarrow$ squats

Rep Counting — State Machine

- Two states (up / down) with per-exercise thresholds
- Counts only on a full down $\rightarrow$ up cycle, avoiding double-counts

Exercise Classifier — CNN

- Learns each exercise from recorded examples
- Labels the exercise automatically — no manual selection
- Smoothed over frames; 98.9% on unseen test data

Core Systems:

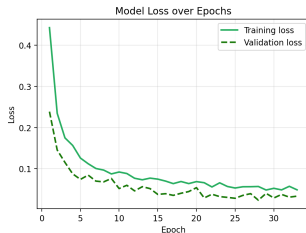
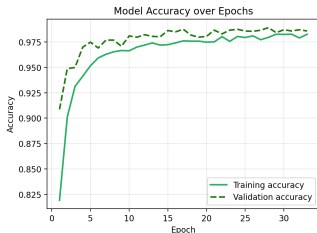
- **Authentication:** Secure local accounts via bcrypt and 6-digit email codes
- **Live Dashboard:** Real-time overlay tracking logs and repetitions concurrently
- **Data History:** Chronological storage logging timestamps and aggregated metrics
- **Email reminders** — daily workout notification at user-defined time via Gmail SMTP

Tech stack:

Component	Technology
UI	CustomTkinter
Pose Estimation	YOLOv8n-Pose
Classifier	TensorFlow CNN
Computer Vision	OpenCV
Database	SQLite
Auth	bcrypt
Email	smtplib + schedule
Language	Python 3.10

CNN Evaluation (1/2)

How the model learns

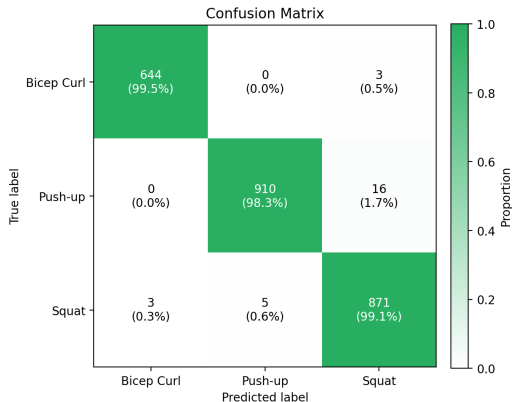


Training behaviour:

- Accuracy **rises** and loss **falls** steadily over the epochs. Training and validation curves stay **close together** — the model **learns well** without overfitting
- Early stopping ends training once validation stops improving

CNN Evaluation (2/2)

How well it classifies each exercise



Confusion matrix:

- Rows = true exercise, columns = predicted
- The **strong green diagonal** means most predictions are correct
- Overall test accuracy: **98.9%**
- The errors that occur are mostly **push-ups vs. curls** — both rely heavily on elbow-angle motion, making them the hardest pair to tell apart

Achieved:

- Real-time performance averaging **134 fps** on an Intel i5-11400H / RTX 3060 laptop
- Automatic classification of 3 exercises with **98.9% accuracy**
- Robust rep counting using joint angle state machines
- Persistent workout history with per-session statistics
- Lightweight by design — no subscription, wearable, or internet required

Future Work

- Expand to 10+ exercise types
- Form quality feedback (e.g. knee caving in squats)
- Mobile version (iOS/Android)
- Cloud sync for multi-device history
- Integration with wearables (HR data)

Thank you!

Pătrașca Andrei Ioan
 West University of Timișoara
 Faculty of Mathematics and Informatics
 Informatics in English

 Questions welcome